

Session 2: Keys, Constraints & Data Integrity

Professional SQL Solutions

Use Case 1: Define Primary Keys in COVID Tables

```
ALTER TABLE covid_cases
ADD CONSTRAINT pk_covid_cases
PRIMARY KEY (country, report_date);
```

Use Case 2: Create Foreign Key Relationships

```
CREATE TABLE countries (
    country VARCHAR(50) PRIMARY KEY,
    population BIGINT
);
```

```
ALTER TABLE covid_cases
ADD CONSTRAINT fk_country
FOREIGN KEY (country)
REFERENCES countries(country);
```

Use Case 3: Add Constraints to Ensure Data Quality

```
ALTER TABLE covid_cases
MODIFY country VARCHAR(50) NOT NULL,
MODIFY report_date DATE NOT NULL,
MODIFY confirmed_cases INT NOT NULL;
```

```
ALTER TABLE covid_cases
ADD CONSTRAINT chk_deaths
CHECK (deaths <= confirmed_cases);
```

Use Case 4: Set Default Values

```
ALTER TABLE covid_cases
MODIFY recoveries INT DEFAULT 0;
```

```
ALTER TABLE covid_cases
ADD COLUMN last_updated TIMESTAMP
DEFAULT CURRENT_TIMESTAMP;
```

Use Case 5: Prevent Duplicate Country-Date Entries

```
ALTER TABLE covid_cases
ADD CONSTRAINT uq_country_date
UNIQUE (country, report_date);
```